



DIAMOND PRICE PREDICTION USING MACHINE LEARNING

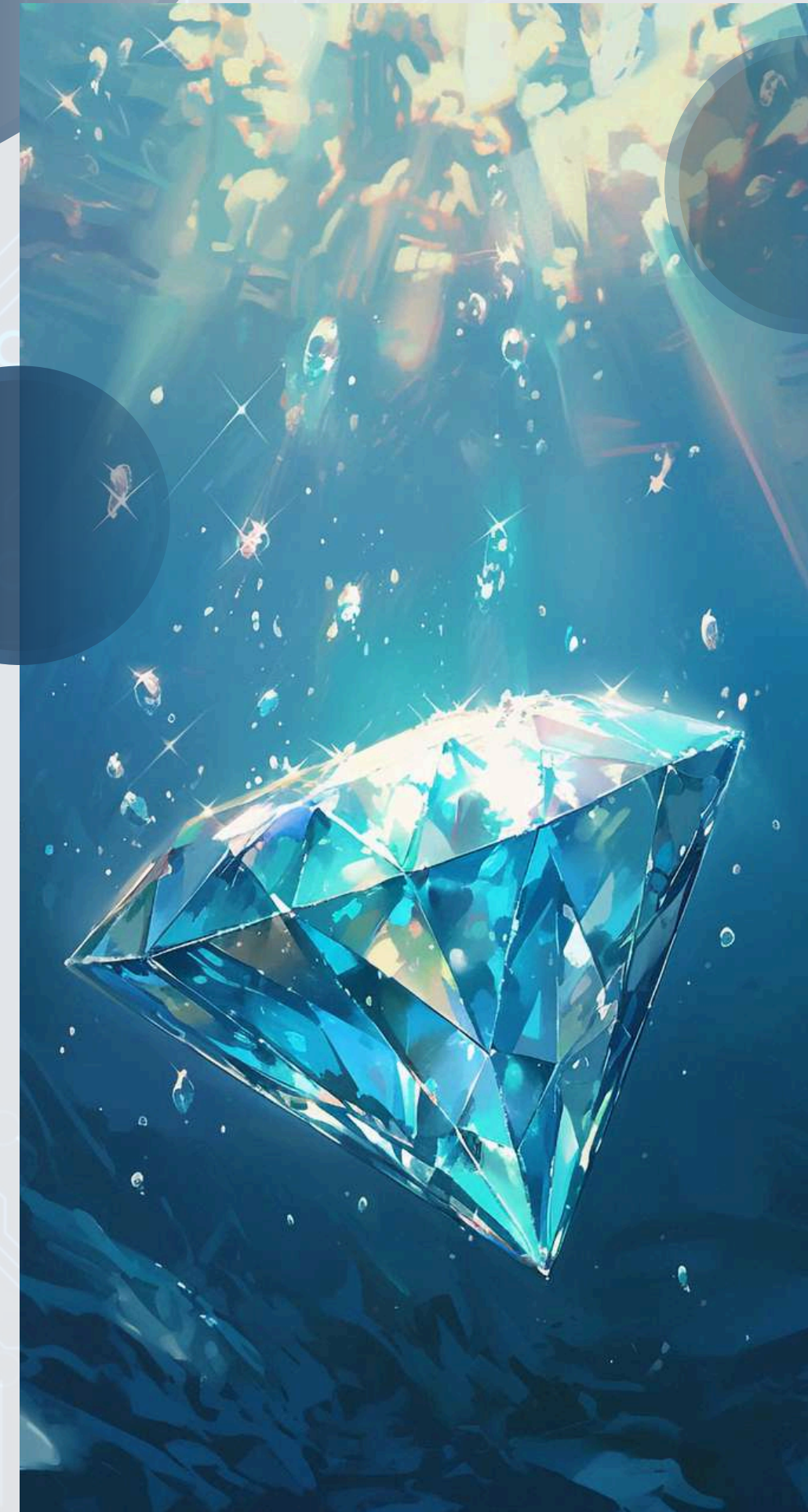


BUSINESS PROBLEM

- ◀ Diamond prices vary depending on several characteristics.
- ◀ Manual valuation is time-consuming.
- ◀ Prices depend on multiple characteristics.
- ◀ Human estimation may be inconsistent.

PROJECT OBJECTIVE

- ◀ Build a regression model to predict the price of a diamond.
- ◀ Compare multiple regression algorithms.
- ◀ Select the model with the best performance.





DATASET OVERVIEW

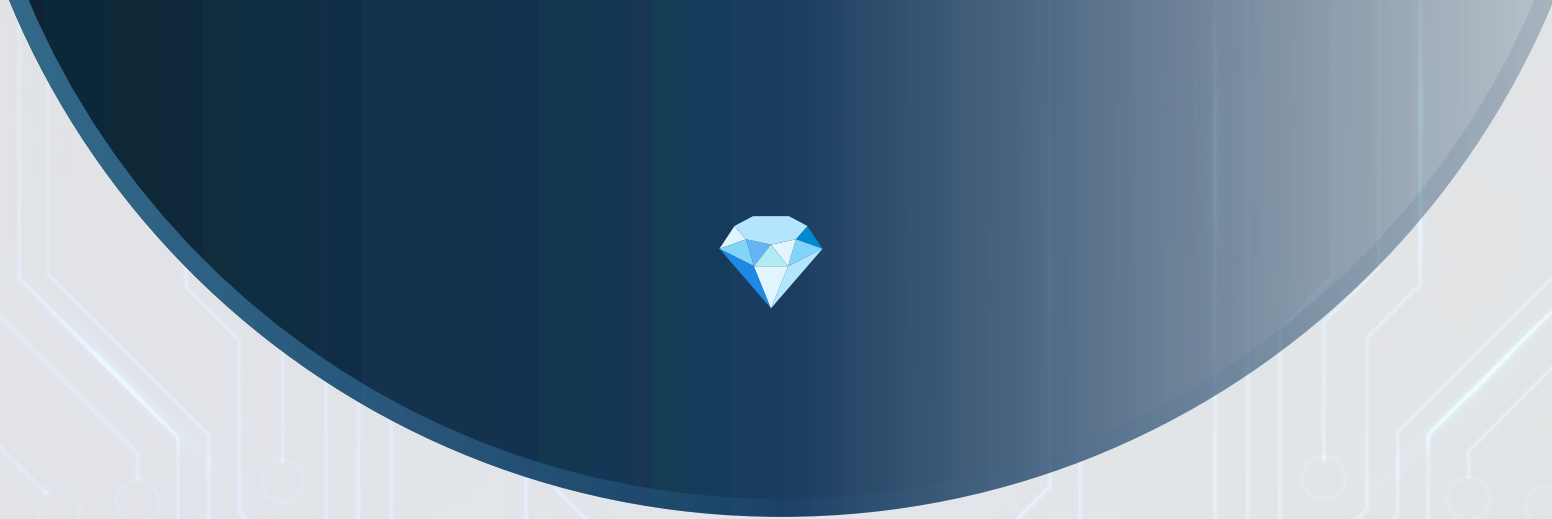
Data set Summary

Property	Value
Dataset	Diamonds
Records	53,940
Features	10
Target	Price

Features

- Numerical** : Carat, Depth, Table, X, Y, Z
- Categorical** : Cut, Color, Clarity
- Target-Variable** : Price

	carat	cut	color	clarity	depth	table	price	x	y	z
0	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
1	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
2	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
3	0.29	Premium	I	VS2	62.4	58.0	334	4.20	4.23	2.63
4	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75



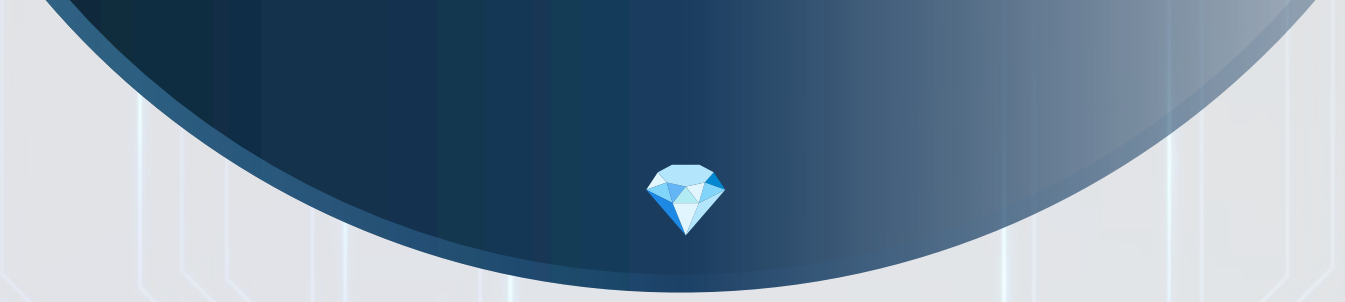
DATA CLEANING & PREPROCESSING

- ◀ Performed preprocessing steps:
 - Checked for missing values
 - Checked for duplicate records
 - Examined outliers
 - Applied Ordinal Encoding
 - Scaled numerical features (where required)
 - Split data into training and testing sets
- ◀ Why Ordinal Encoding?
 - Cut, Color, and Clarity have a natural order.

Example:

Fair → Good → Very Good → Premium → Ideal





EXPLORATORY DATA ANALYSIS

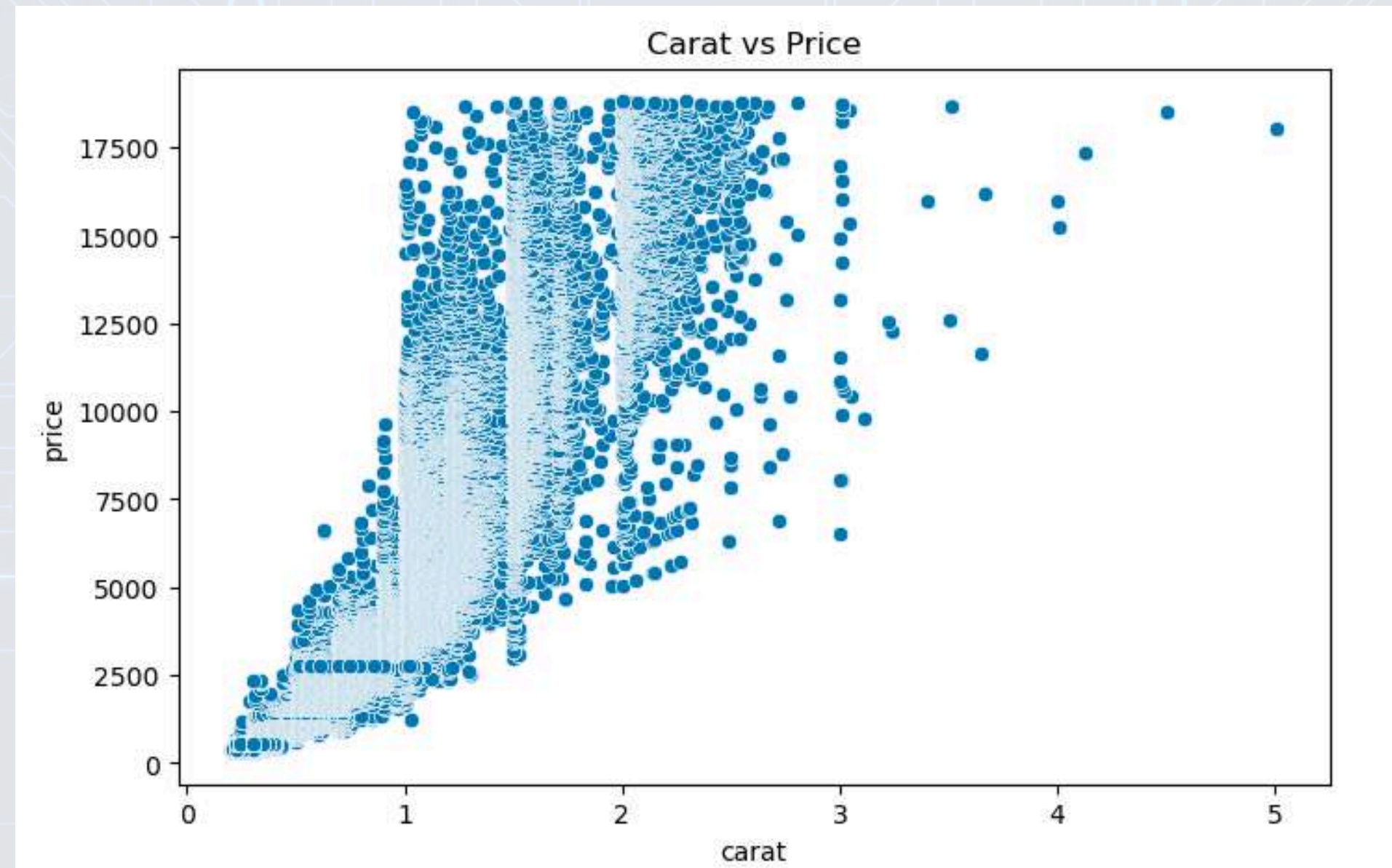
Visualizations used

- Histogram
- Boxplot
- Scatter Plot
- Correlation Heatmap
- Distribution Plot

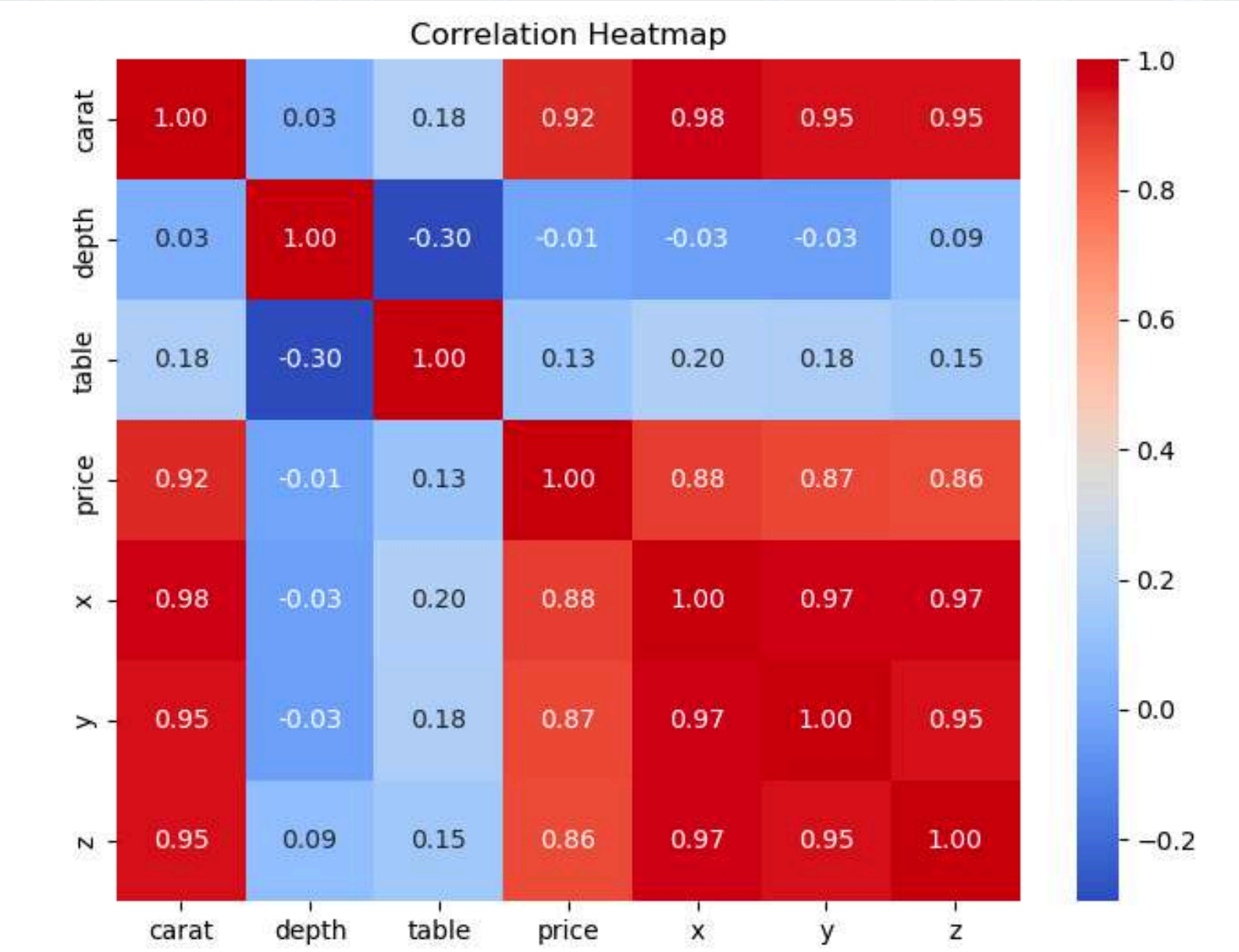
Key Findings

- Carat has a strong positive relationship with price.
- x, y, and z are highly correlated.
- Outliers exist in price and carat.

SCATTER PLOT (CARAT VS PRICE)



CORRELATION HEATMAP



FEATURE ENGINEERING

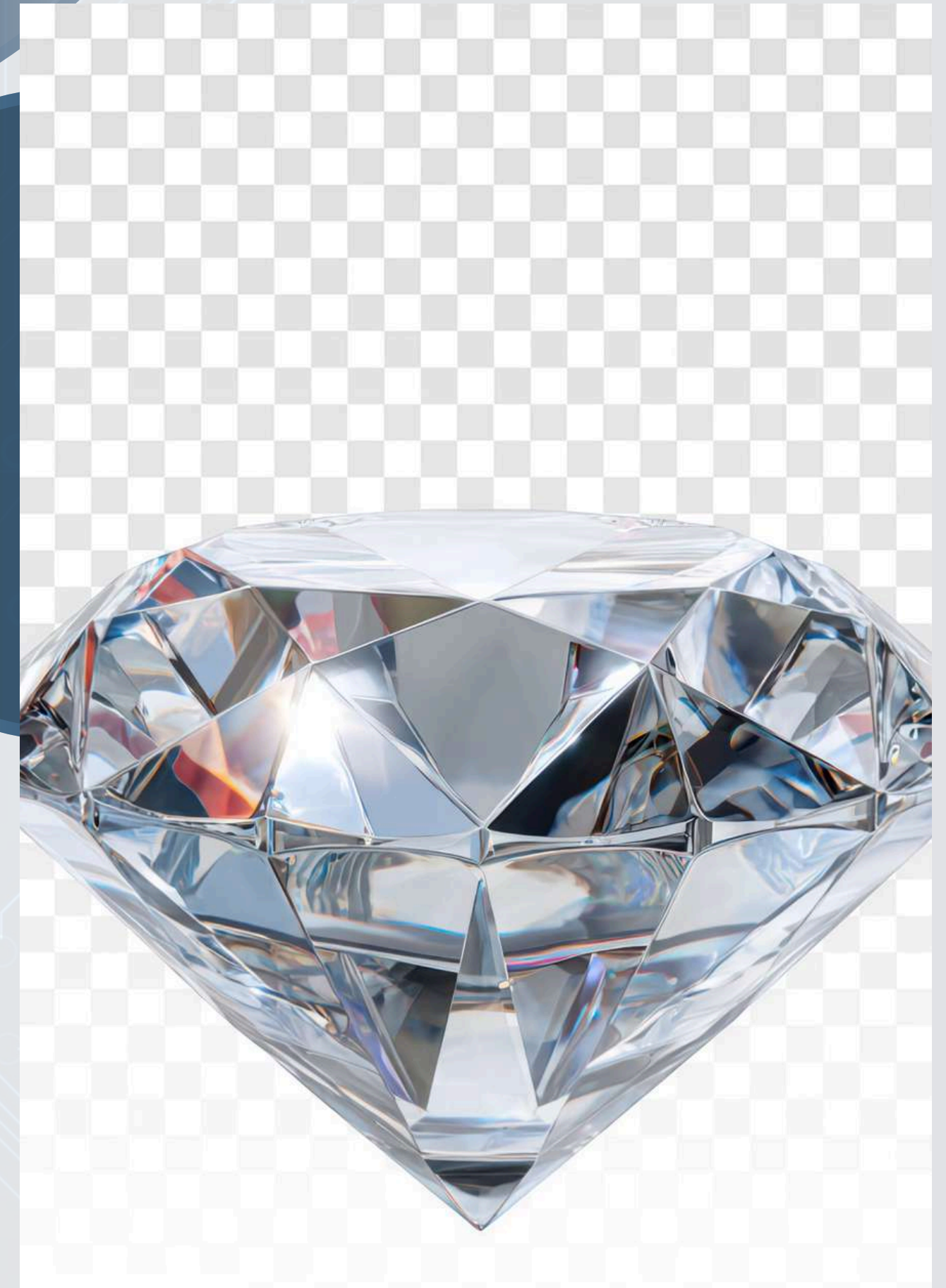
- No additional features were created.
- Existing features already provided sufficient information.
- Models achieved excellent performance without creating new variables.

Optional Feature

- $\text{Volume} = x \times y \times z$

Reason for not using it

- Original features were already effective.



REGRESSION ALGORITHMS USED

- ◀ Linear Regression
- ◀ Ridge Regression
- ◀ Lasso Regression
- ◀ Decision Tree
- ◀ Random Forest
- ◀ Gradient Boosting

- ◀ SVR
- ◀ AdaBoost
- ◀ Extra Trees
- ◀ XGBoost
- ◀ LightGBM
- ◀ CatBoost



MODEL EVALUATION

Metric

MAE
MSE
RMSE
 R^2 Score

Purpose

Average prediction error
Measures average squared error
Average error (same unit as price)
Explains variation in the data

Better Value

Lower
Lower
Lower
Higher (closer to 1)



MODEL COMPARISON

We compared all trained models using the same evaluation metrics to identify the best performer.

	Algorithm	Training Time (s)	MAE	MSE	RMSE	R ² Score
0	Linear Regression	1.380244	858.708470	1.825913e+06	1351.263480	0.885140
1	Ridge Regression	0.663074	858.800082	1.825909e+06	1351.262011	0.885140
2	Lasso Regression	0.576920	859.684670	1.826229e+06	1351.380538	0.885120
3	Decision Tree	1.147414	355.644234	5.314899e+05	729.033563	0.966566
4	Random Forest	35.384832	267.975502	2.938155e+05	542.047468	0.981517
5	Gradient Boosting	8.716423	364.865262	4.299356e+05	655.694773	0.972955
6	Support Vector Regression	248.737321	1346.839299	7.717448e+06	2778.029487	0.514529
7	Extra Trees	27.444980	263.968500	2.953158e+05	543.429690	0.981423
8	AdaBoost	8.731816	1296.366021	2.360226e+06	1536.302559	0.851528
9	XGBoost	5.761056	276.373871	2.778682e+05	527.132070	0.982521
10	LightGBM	0.651659	284.858833	2.879718e+05	536.630074	0.981885
11	CatBoost	8.787783	316.758435	3.380345e+05	581.407350	0.978736

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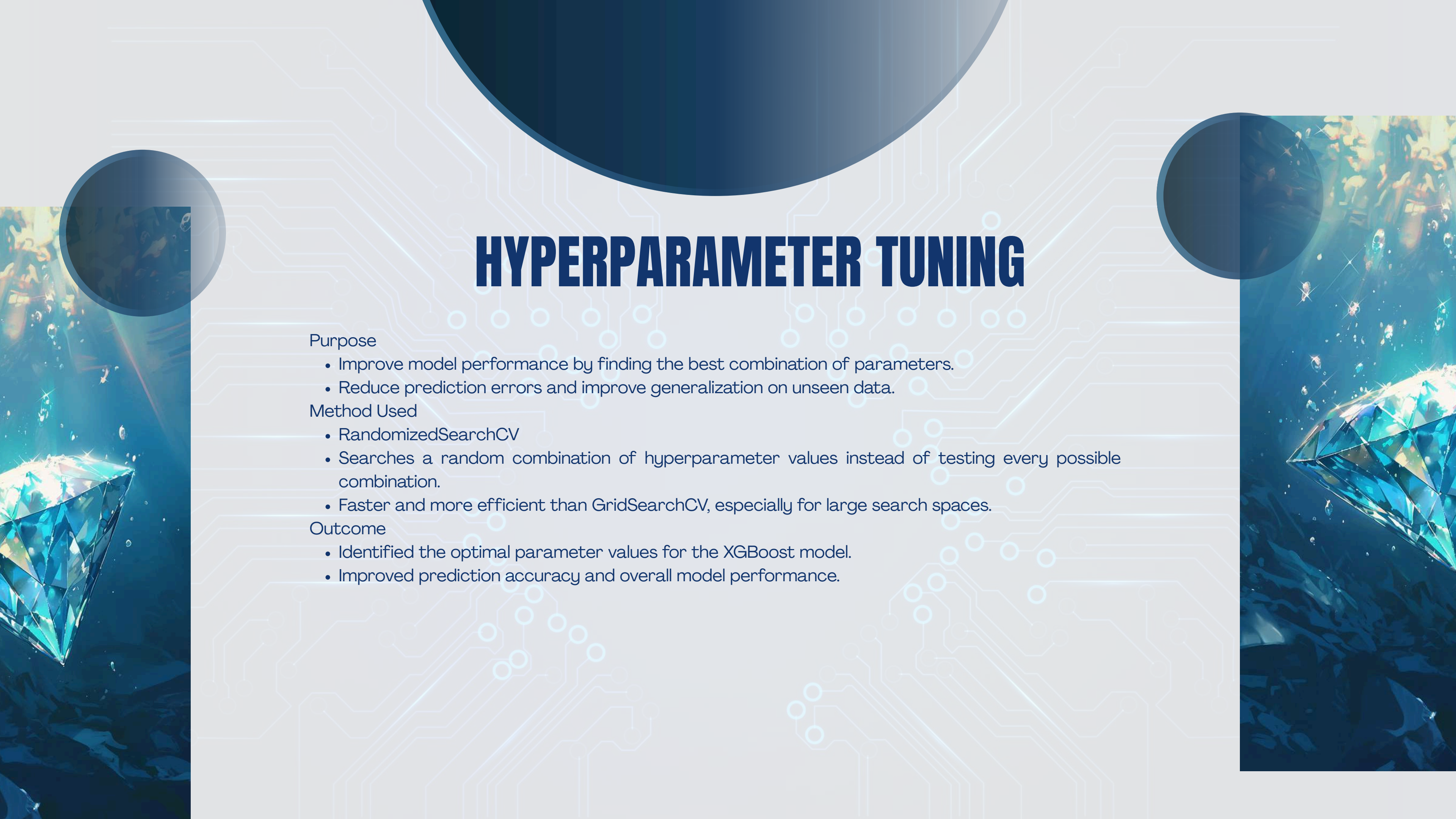
BEST MODEL SELECTION

Best Model: XGBoost Regression

Why XGBoost?

- Achieved the highest R^2 score, indicating the best fit to the data.
- Produced the lowest prediction error compared to the other regression models.
- Generalized well to unseen test data.
- Handles complex, non-linear relationships effectively.
- Includes regularization techniques that help reduce overfitting.





HYPERPARAMETER TUNING

Purpose

- Improve model performance by finding the best combination of parameters.
- Reduce prediction errors and improve generalization on unseen data.

Method Used

- RandomizedSearchCV
- Searches a random combination of hyperparameter values instead of testing every possible combination.
- Faster and more efficient than GridSearchCV, especially for large search spaces.

Outcome

- Identified the optimal parameter values for the XGBoost model.
- Improved prediction accuracy and overall model performance.

LIVE PREDICTION DEMO

- ◀ The trained model was integrated into a Streamlit application, allowing users to enter diamond characteristics and receive an instant price prediction.

Technologies Used

- Streamlit – Used to build the web interface.
- Joblib – Used to load the trained machine learning model.
- XGBoost – Performs the price prediction.
- Pandas & NumPy – Handle and prepare the input data.

The screenshot shows a web application titled "Diamond Price Prediction System". It features a sidebar on the left with navigation links: "app", "About Team", "Dataset Information", "Model Comparison", "Model Information", "Predict Form", "Prediction Result", and "Project Description". The main content area has a header "Diamond Price Prediction System" with a diamond icon. Below it is a subtitle "Predict Diamond Prices Using Machine Learning" and a description: "This web application predicts the price of a diamond using an optimized XGBoost Regression Model trained on the Diamonds dataset. Navigate through the pages using the sidebar to learn about the project, explore the dataset, compare models, and predict diamond prices." A table displays key metrics: Dataset Size (53,940), Features (10), Models Tested (12), and Best Model (XGBoost). Below the table is a section titled "Project Highlights" with two items: "✓ Data Cleaning" and "✓ 12 Regression Models".

Dataset Size	Features	Models Tested	Best Model
53,940	10	12	XGBoost

Project Highlights

- ✓ Data Cleaning
- ✓ 12 Regression Models

A woman's face is shown in profile, looking towards the right. Her face is semi-transparent, revealing a complex circuit board pattern underneath. The circuit board has various components like chips, resistors, and traces. The background is a dark blue gradient with a large, faint circular arc on the right side. On the left, there are some glowing white dots and lines, suggesting a network or data flow.

THANK YOU.